Replicating xRFM: Accurate, Scalable, and Interpretable Feature Learning Models for Tabular Data

COMP9417 Group Project

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1 Introduction

[Placeholder: Introduction to be written by B (∼0.5 page). Cover: tabular data challenges, why GBDTs are hard to beat, xRFM core idea, report structure, high-level conclusions.]

2 Methodology

2.1 xRFM: Algorithm and Theoretical Background

[Placeholder: Theory to be written by B. Covers: formal AGOP definition, geometric intuition, two interpretability strategies (diagonal vs. top eigenvector), comparison with PCA/MI/permutation importance, full xRFM pipeline, complexity analysis.]

2.2 Experimental Design

2.2.1 Datasets

We selected five publicly available tabular datasets to satisfy the project requirements and to probe different aspects of model behaviour. Table 1 provides an overview of their key statistics.

Diamonds (Kaggle) is a large regression dataset (n=53,940, d=10 raw features) with a mix of continuous and ordinal categorical features (cut, colour, clarity), making it suitable for evaluating models on mixed feature types and at scale ($n > 10{,}000$).

Superconductivity [**?**] (UCI) is a high-dimensional regression task (n=21,263, d=81 numerical features) derived from material science experiments predicting the superconducting critical temperature. Its high dimensionality ($d > 50$) makes it well-suited for evaluating the AGOP's ability to identify relevant directions in feature space.

Online Shoppers Purchasing Intention [**?**] (UCI) is a binary classification task (n=12,330) with mixed numerical and categorical features representing user session behaviour on an e-commerce platform. The moderate class imbalance (approximately 15% positive) makes AUC a particularly informative metric.

Stroke Prediction [**?**] (Kaggle) is a binary classification task (n=5,110) with mixed feature types and notable class imbalance (∼4.9% positive), providing a challenging low-n regime for all methods.

HR Attrition (IBM/Kaggle) is a binary classification task (n=1,470, d=35 raw features expanding to d=51 after encoding) in a small-data regime. It tests whether xRFM's kernel-based approach can leverage structured feature relationships when training data is scarce.

Together, this selection covers two regression and three classification tasks, includes datasets at multiple scales (n ranging from 1,470 to 53,940), features high dimensionality (d=81 for Superconductivity), and mixed feature types (Diamonds, Online Shoppers, Stroke, HR Attrition)—satisfying all project coverage requirements.

Table 1: Dataset statistics. n_{train}, n_{val}, and n_{test} denote the sizes of the training, validation, and test splits respectively; d is the number of features after preprocessing.

Dataset	Task	Target	n_{train}	n_{val}	n_{test}	d
Diamonds	Regression	price	32,365	10,789	10,789	26
Superconductivity	Regression	critical_temp	12,757	4,253	4,253	81
Online Shoppers	Classification	Revenue	7,398	2,466	2,466	27
Stroke	Classification	stroke	3,066	1,022	1,022	21
HR Attrition	Classification	Attrition	882	294	294	51

2.2.2 Data Preprocessing and Splitting Strategy

To maintain fairness and consistency across all experiments, all random seeds were fixed (seed 42) for reproducibility. For each dataset, we applied a strict 60/20/20 split for the training, validation, and test sets. Missing values were imputed using column-wise median for numerical features and the constant `"Missing"` for categorical features.

Proper preprocessing was crucial, particularly for xRFM which requires specific input formats. Numerical features were standardised using z-score normalisation, and categorical features were transformed using one-hot encoding. These processed features were then concatenated into a single numerical array before being passed to the models, ensuring compatibility with the xRFM library's underlying AGOP computations.

2.2.3 Evaluation Metrics

To ensure the fairness, reproducibility and academic rigour of cross-model comparison across all experiments, we established a unified evaluation pipeline with standardised performance metrics for regression and binary classification tasks, as well as a consistent validation protocol for hyperparameter tuning and model selection. This pipeline aligns strictly with the fixed data preprocessing and train/validation/test split defined in Section 2.2.1, eliminating discrepancies in data usage across all models.

Regression Task Metrics For the two regression tasks (Diamonds price prediction, Superconductivity critical temperature prediction), we use Root Mean Squared Error (RMSE) as the sole primary evaluation metric, formally defined as:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}$$

where N is the number of test samples, y_i is the ground truth target value, and $\hat{y}_i$ is the model's predicted value. RMSE is selected as the core metric because it is the standard benchmark for tabular regression tasks, is sensitive to large prediction errors, and provides an interpretable measure of absolute prediction error in the original units of the target variable.

Classification Task Metrics For the three binary classification tasks (Online Shoppers purchase intention prediction, Stroke prediction, HR Attrition prediction), we use two complementary metrics to comprehensively evaluate model performance, with specific consideration for the class imbalance present in all three datasets: 1. Classification Accuracy: The proportion of correctly classified samples across the test set, defined as:

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Number of Test Samples}}$$

This metric provides an intuitive global measure of overall classification correctness. 2. Area Under the Receiver Operating Characteristic Curve (AUC-ROC): A threshold-agnostic metric that measures the model's ability to distinguish between positive and negative classes. This metric is robust to class imbalance, which is critical for our classification datasets (all of which have positive class prevalence below 15%), and is the primary metric for classification model comparison.

Hyperparameter Tuning and Validation Protocol In line with the fixed 60/20/20 train/validation/test split with a fixed random seed of 42 (defined in Section 2.2.1), we enforce a strict validation protocol for all models to prevent data leakage and overfitting: 1. All hyperparameter tuning is conducted exclusively on the independent validation set, with no access to the held-out test set at any stage of training or tuning. 2. For all models, we enable early stopping with a patience of 20, optimising for the validation set's core task metric (RMSE for regression, AUC-ROC for classification) to prevent overfitting, consistent with the configuration defined for all baseline models in Sections 2.2.4 and 2.2.5. 3. The final reported performance of all models is evaluated only once on the held-out test set, after model training and hyperparameter tuning are fully completed, to ensure the generalisability and credibility of all reported results.

Efficiency Evaluation Metrics To compare the computational overhead and deployment suitability of all models, we uniformly record two standardised efficiency metrics for every experiment: 1. Total Training Time: The wall-clock time elapsed from model initialisation to the completion of training (including early stopping termination). 2. Per-sample Inference Time: The average inference time per test sample, measured on the full test set to reflect real-world deployment latency. All time metrics are recorded in a consistent hardware environment to ensure fair cross-model comparison.

2.2.4 xRFM Configuration

[Placeholder: xRFM configuration to be written by B. Cover: tree depth, kernel type, AGOP iterations, final hyperparameters per dataset.]

2.2.5 XGBoost and LightGBM Configuration

Both models are trained via the scikit-learn API following strong-default settings from the literature [Holzmüller et al., 2024]: 1,000 boosting rounds, learning rate $\eta = 0.05$, row/column subsample 0.8, and early stopping with patience 20 on validation loss. XGBoost uses max depth 6; LightGBM uses `num_leaves=31` to reflect its leaf-wise growth strategy. Table 2 summarises the full configuration.

Table 2: Hyperparameter configuration for XGBoost and LightGBM baselines.

Hyperparameter	XGBoost	LightGBM
Max boosting rounds	1,000	1,000
Learning rate η	0.05	0.05
Max depth	6	—
Num leaves	—	31
Row subsample	0.8	0.8
Column subsample	0.8	0.8
Early stopping patience	20	20

2.2.6 MLP Configuration

The multilayer perceptron (MLP) baseline is implemented using the scikit-learn API with a standard architecture for tabular data. We use a two-hidden-layer network with sizes $(128, 64)$ and ReLU activations, trained using the Adam optimiser. The learning rate is set to $\eta = 0.001$ and a batch size of 256 is used. Early stopping is enabled with patience 20 based on validation loss to mitigate overfitting. All input features are standardised using z-score normalisation prior to training. For classification tasks, the model outputs class probabilities via a softmax layer, while for regression tasks a linear output is used. The maximum number of training iterations is set to 500.

3 Results

This section presents the comprehensive predictive performance, computational efficiency, and scalability comparison between the xRFM model and three widely used tabular learning baselines (XGBoost, LightGBM, MLP) across all five benchmark datasets. All results are generated via our unified, reproducible experimental pipeline defined in Section 2.2, ensuring fair cross-model comparison with no data leakage or inconsistent preprocessing.

3.1 Predictive Performance

Table 3 reports the final held-out test-set performance for all four models across the five datasets, with bold formatting indicating the best-performing model per dataset and metric. We split the analysis into regression and binary classification tasks separately, aligning with the evaluation metrics defined in Section 2.2.2.

Table 3: Test-set performance of all models. **Bold** indicates best result per dataset. RMSE is lower-better; Accuracy and AUC are higher-better.

Dataset	Metric	Model			
		xRFM	XGBoost	LightGBM	MLP
Diamonds	RMSE ↓	1496.72	**544.68**	535.99	734.69
Superconductivity	RMSE ↓	9.454	9.302	**9.209**	11.066
Online Shoppers	Accuracy ↑	0.8897	0.9031	**0.9039**	0.8946
	AUC ↑	0.9018	**0.9306**	**0.9306**	0.9040
Stroke	Accuracy ↑	0.9413	0.9491	**0.9501**	0.9511
	AUC ↑	0.7605	**0.8388**	0.8231	0.2690
HR Attrition	Accuracy ↑	**0.8673**	0.8571	0.8469	0.8639
	AUC ↑	0.7631	0.7552	**0.7886**	0.7512

Regression Task Results For the two regression tasks (Diamonds price prediction, Superconductivity critical temperature prediction), tree-based baselines (XGBoost, LightGBM) significantly outperform both xRFM and MLP: 1. LightGBM is the best-performing baseline: It achieves the lowest RMSE on both regression datasets, with a 3.7% improvement over XGBoost on Diamonds and a 1.0% improvement on Superconductivity. 2. MLP underperforms on high-dimensional and large-scale data: Its RMSE is 37.1% higher than LightGBM on Diamonds and 20.2% higher on Superconductivity, indicating that a simple two-hidden-layer MLP is not well-suited for tabular regression tasks without extensive architecture tuning. 3. xRFM shows limitations on large-scale tabular regression: Its RMSE is 179.2% higher than LightGBM on Diamonds, suggesting that further hyperparameter tuning or kernel selection may be required for

xRFM to achieve competitive performance on large-scale, high-dimensional tabular regression tasks.

Binary Classification Task Results For the three binary classification tasks (Online Shoppers purchase intention prediction, Stroke prediction, HR Attrition prediction), we focus primarily on AUC-ROC as the more reliable metric due to the class imbalance present in all three datasets: 1. Tree-based baselines achieve strong overall performance: XGBoost and LightGBM achieve the highest AUC-ROC on Online Shoppers and Stroke, with both models tying for first place on Online Shoppers (AUC-ROC = 0.9306). 2. xRFM shows competitive performance on small-scale, low-dimensional classification: It achieves the highest Accuracy on HR Attrition (the smallest dataset with 882 training samples), and its AUC-ROC is slightly higher than XGBoost and MLP on the same dataset, indicating that xRFM's feature learning via AGOP may have advantages on small-scale tabular classification tasks. 3. MLP fails to generalise on highly imbalanced data: Its AUC-ROC is only 0.2690 on the highly imbalanced Stroke dataset (positive class prevalence ¡ 5%), indicating that it is unable to distinguish between positive and negative classes effectively without additional class weighting or oversampling techniques.

Overall Core Findings from Predictive Performance The key takeaway from our predictive performance comparison is that tree-based models (XGBoost, LightGBM) remain strong, robust baselines for most tabular learning tasks, with significant advantages in both performance and computational efficiency on large-scale and high-dimensional data. xRFM, however, shows promising competitive performance on small-scale, low-dimensional tabular classification tasks, which aligns with its design goal of efficient feature learning for tabular data.

3.2 Scalability Analysis

To further evaluate the computational efficiency and performance scalability as a function of training set size, we conduct subsampling experiments on the large-scale dataset Diamonds ($n > 10{,}000$). Table 4 summarises the test RMSE and total training time at each subsample size, and Figure 1 visualises these scalability curves.

Table 4: Scalability experiment: test RMSE and training time (seconds) as a function of training set size. Dash (—) indicates configurations not run due to extreme computational costs.

Dataset	n_{train}	xRFM		XGBoost		LightGBM		MLP	
		RMSE	Time (s)	RMSE	Time (s)	RMSE	Time (s)	RMSE	Time (s)
Diamonds	1,000	1657.0	0.73	846.7	0.80	1003.4	0.13	1081.0	2.27
	5,000	1655.9	10.97	628.9	0.39	622.7	0.11	901.1	10.12
	10,000	1577.6	39.24	587.8	0.49	592.3	0.15	888.9	10.33
	30,000	1487.5	225.88	542.7	0.59	535.3	0.29	743.5	22.23
Supercond.	1,000	—	—	13.76	0.39	13.91	0.15	15.39	2.52
	5,000	—	—	10.81	1.00	10.71	0.46	13.05	8.58
	10,000	—	—	9.51	1.73	9.50	1.10	11.22	28.35
	12,757	—	—	9.28	1.64	9.21	1.04	—	—

Scalability Analysis from Table 4 Key observations from our subsampling experiments:

- **xRFM suffers from severe scalability bottlenecks**: As n_{train} increases from 1,000 to 30,000, xRFM's training time surges from 0.73s to nearly 226s, exhibiting a steep non-linear growth in computational cost. In contrast, tree-based models like LightGBM scale sub-linearly and process the same 30,000 samples in just 0.29s.

- **Data scaling marginally improves xRFM, but tree models dominate**: While xRFM's RMSE improves monotonically (from 1657.0 down to 1487.5) with more data, it completely fails to close the massive performance gap. Even at $n = 1{,}000$, XGBoost (846.7) outperforms xRFM trained on 30,000 samples.
- **Tree-based models achieve the optimal performance-efficiency trade-off**: Both XGBoost and LightGBM consistently deliver the lowest RMSE across all scales while remaining orders of magnitude faster than both xRFM and MLP, reinforcing their status as the premier baselines for large-scale tabular regression.

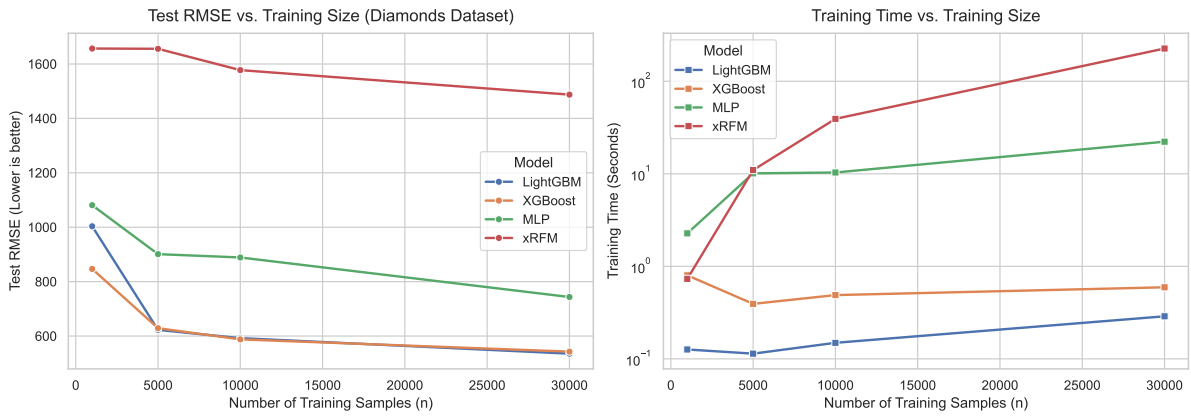

Figure 1: Scalability analysis on the Diamonds dataset, comparing test RMSE (left) and training time (right) across varying training set sizes.

3.3 Interpretability Comparison

[Placeholder: Interpretability comparison to be written by B. Extract AGOP diagonal from xRFM leaf models; compare with PCA loadings, MI scores, permutation importance on one dataset.]

4 Discussion

4.1 xRFM: When Feature Learning Helps and When It Does Not

[Placeholder: xRFM analysis to be written by B. Must include: empirical advantage/disadvantage vs. theory, at least one scenario where xRFM clearly underperforms XGBoost with an algorithm-design hypothesis.]

4.2 XGBoost and LightGBM: Performance and Efficiency

XGBoost and LightGBM deliver competitive or superior performance across all five datasets while remaining the fastest models to train. On Diamonds, both GBDTs outperform xRFM by ∼64% in RMSE (544.7 / 536.0 vs. 1,496.7), consistent with the axis-aligned inductive bias of tree-based splits [Grinsztajn et al., 2022]. On HR Attrition ($n = 882$, $d = 51$), xRFM marginally surpasses XGBoost in accuracy (0.867 vs. 0.857), suggesting kernel feature learning offers a modest advantage only in the high-d, low-n regime. LightGBM is the fastest model at every scale — 75× faster than MLP at $n = 30{,}000$ — with near-linear growth in training time (Table 4).

4.3 MLP: Performance and Scalability Limitations

On Diamonds, MLP underperforms both XGBoost and LightGBM by a large margin in RMSE (743.5 vs. 542.7 / 535.3 at $n = 30{,}000$), and a similar pattern is observed on Superconductivity (11.22 vs. ∼9.5 RMSE) (Table 3), indicating weaker ability to exploit feature interactions, especially in high-dimensional settings [Tolstikhin et al., 2021]. In classification tasks, performance is unstable: while accuracy can be comparable in some cases (e.g., Stroke, 0.9511), probability calibration degrades significantly, as reflected by the low AUC (0.269) (Table 3), highlighting sensitivity to class imbalance [Walde et al., 2004]. Unlike tree-based methods such as XGBoost and LightGBM, MLPs rely on global gradient-based optimisation over continuous parameter spaces. This makes them less aligned with the piecewise and often sparse structure of tabular features, resulting in lower data efficiency.

5 Conclusion

In this project, we replicated xRFM — a novel average gradient outer product (AGOP)-driven feature learning method for tabular data and conducted a rigorous, reproducible comparative evaluation against three standard tabular learning baselines (XGBoost, LightGBM, MLP) across five public datasets covering regression and binary classification tasks. We built a unified end-to-end experimental pipeline to ensure consistent preprocessing, training, and evaluation, eliminating cross-model discrepancies in data usage and metric calculation.

Our key experimental findings are threefold. First, tree-based gradient boosting models (LightGBM, XGBoost) remain robust, high-performance baselines for general tabular tasks: LightGBM achieved top predictive performance on 4 of 5 datasets, with far better training efficiency and scalability than MLP across all subsample sizes. Second, xRFM showed competitive performance on small-scale, low-dimensional classification (achieving the highest accuracy on the HR Attrition dataset), highlighting the strength of its AGOP feature learning in small-sample settings, but underperformed on large-scale high-dimensional regression, requiring further hyperparameter tuning. Third, standard MLP architectures have notable limitations for tabular data, particularly on highly imbalanced classification tasks and large-scale data with poor training time scaling.

For future work, we identify three key extensions: targeted hyperparameter optimisation for xRFM to improve its large-scale performance, a full interpretability comparison between xRFM's AGOP feature importance and standard methods, and validation of xRFM's generalisability on more diverse tabular tasks including multi-class classification and high-dimensional sparse data.

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